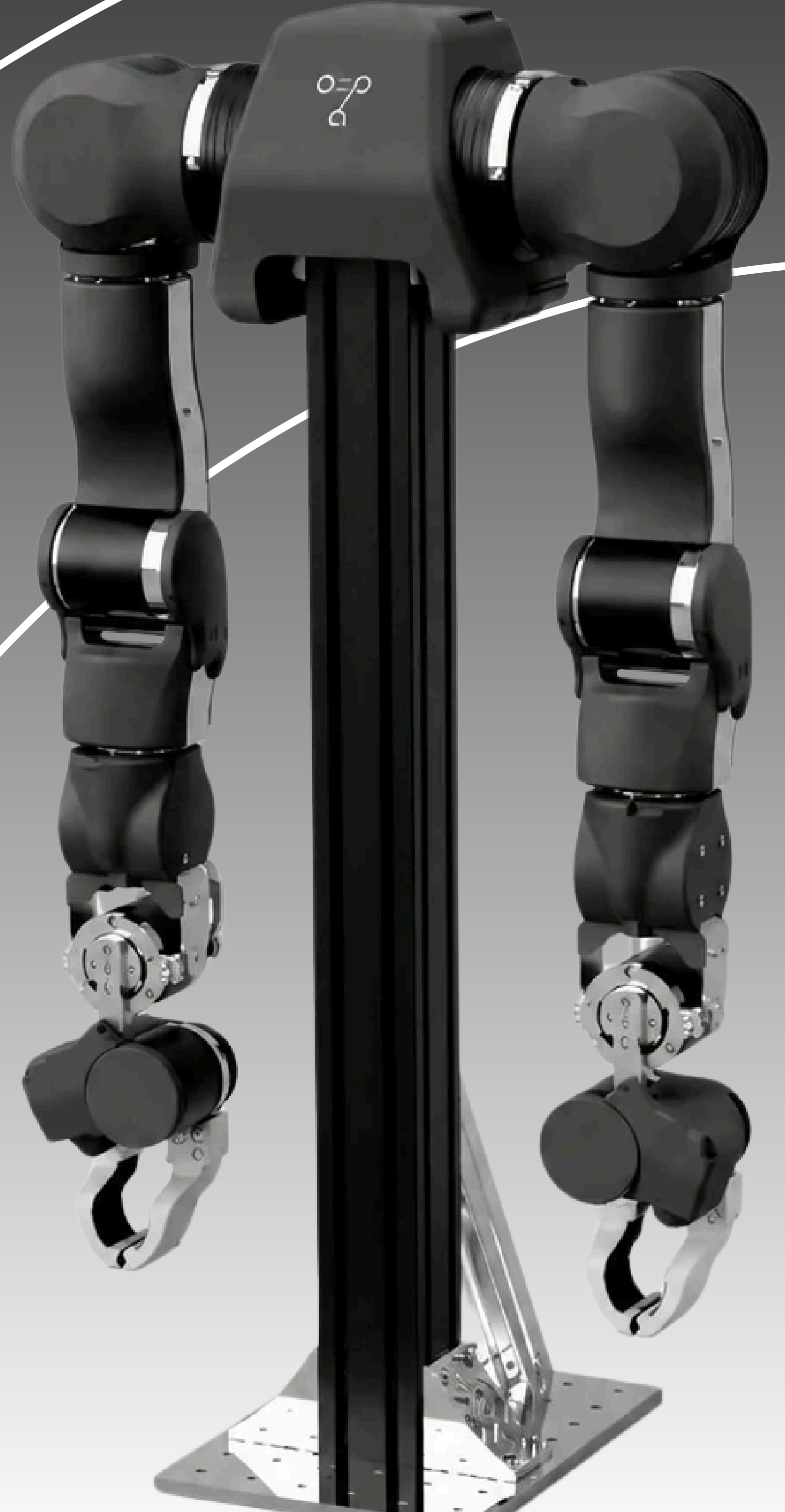
Reagan Raphael

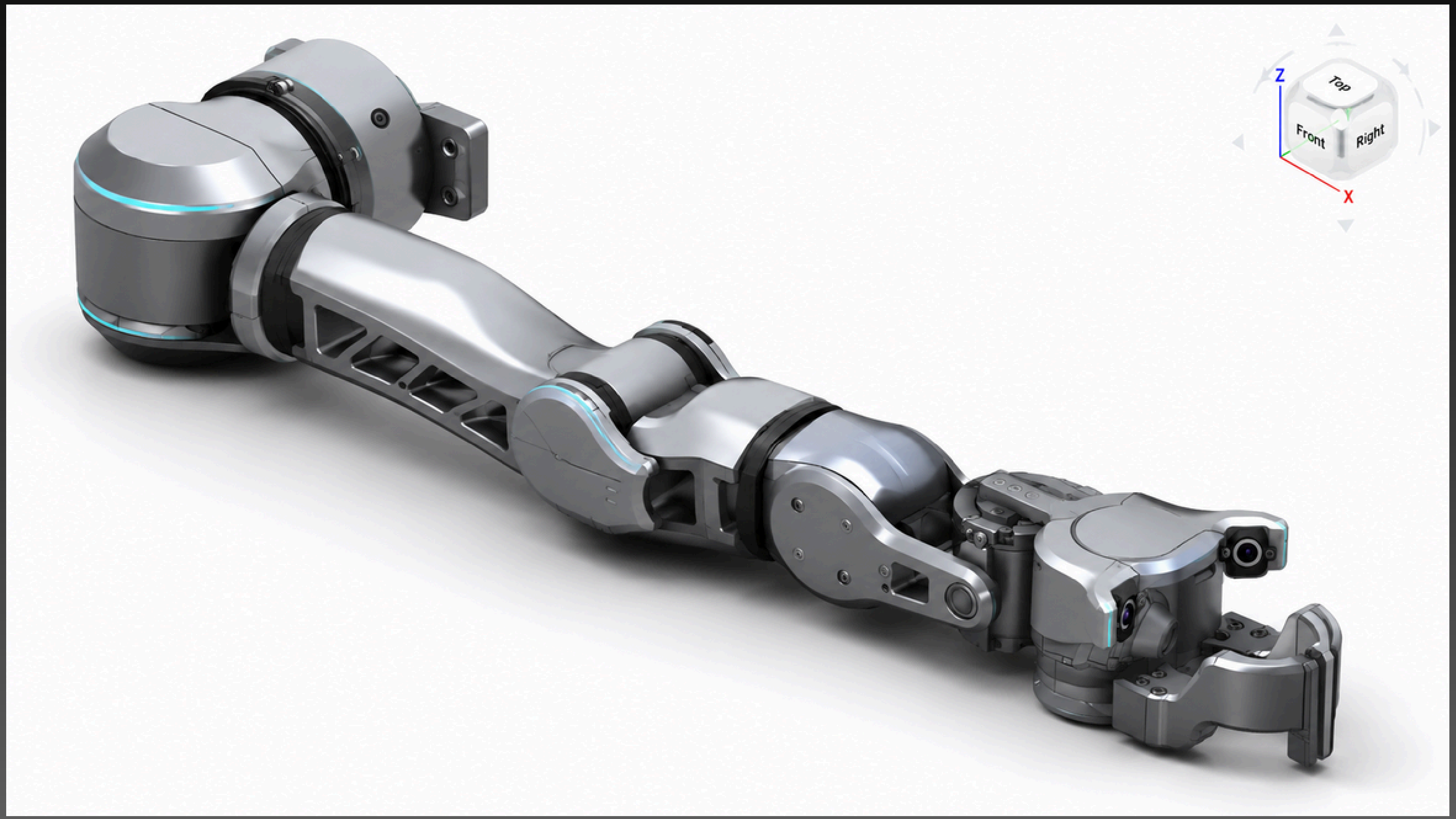
OpenArm 2.0 housing design

DeepAware AI / SVRC

June 2026

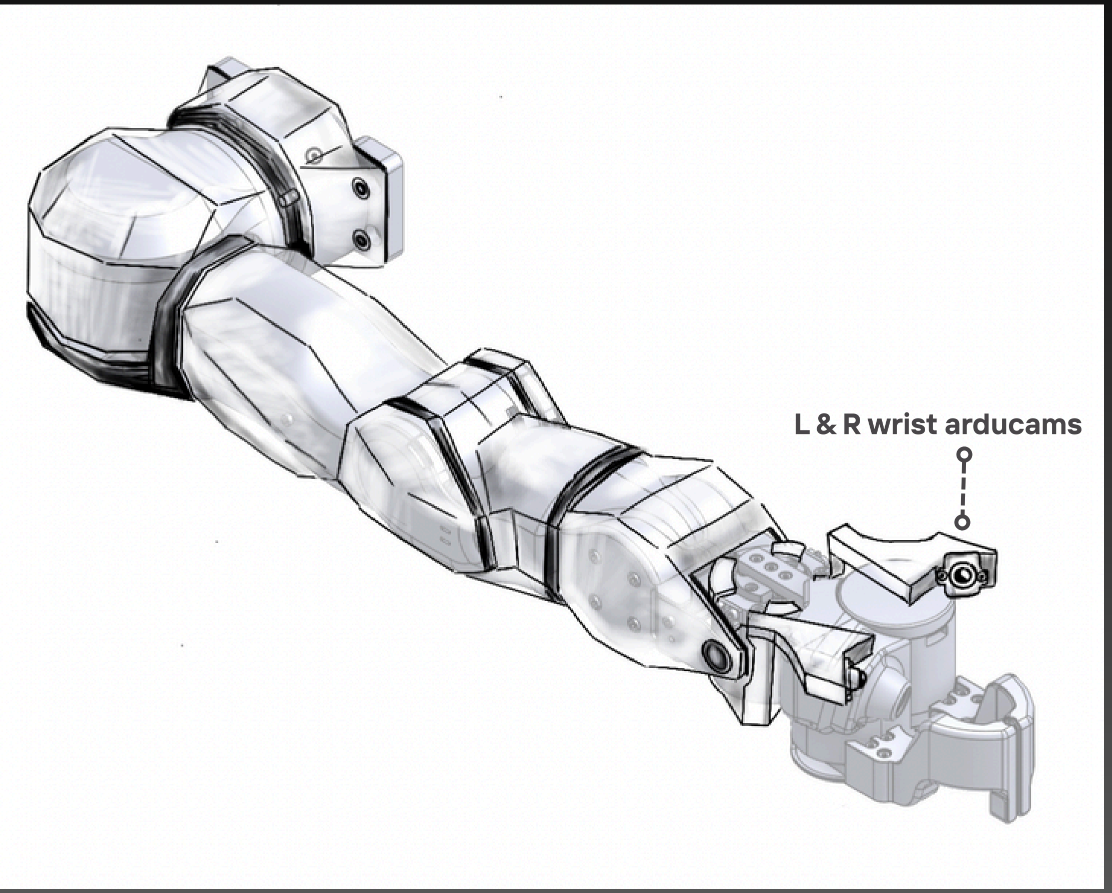


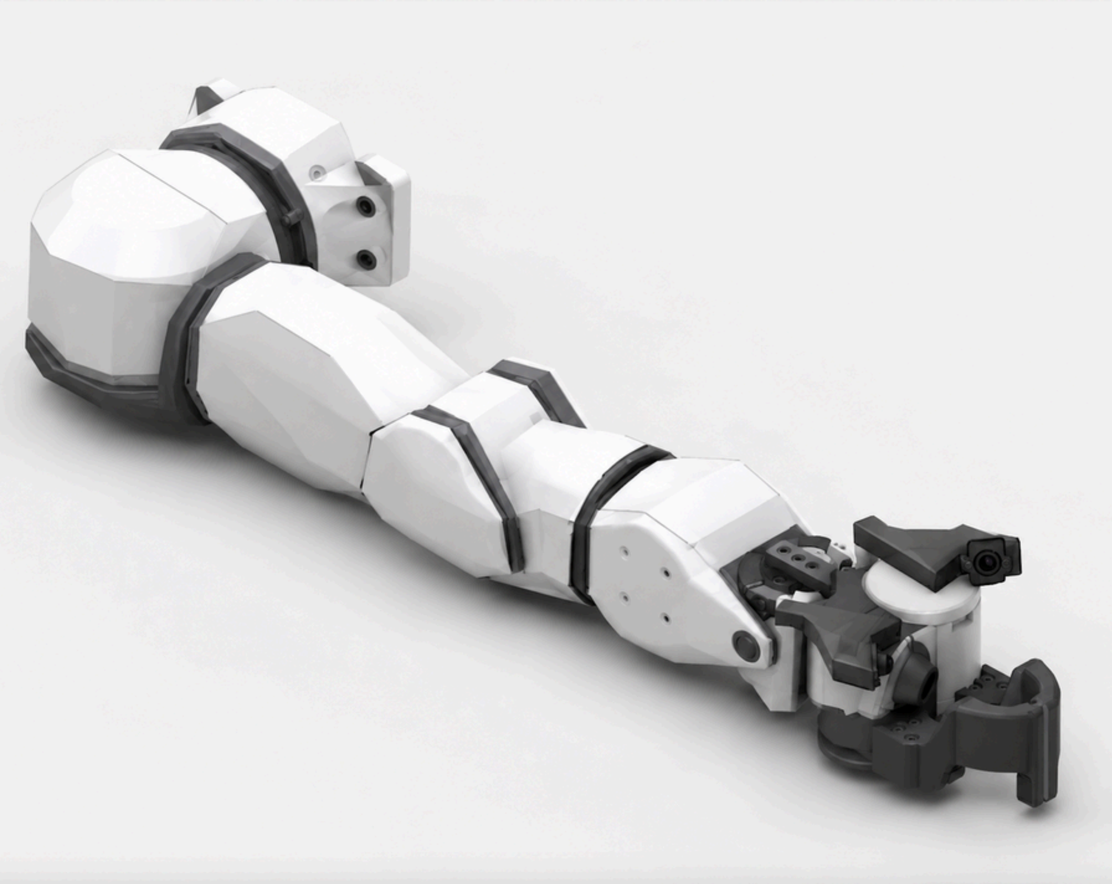
Design Direction 1: Industrial Silver



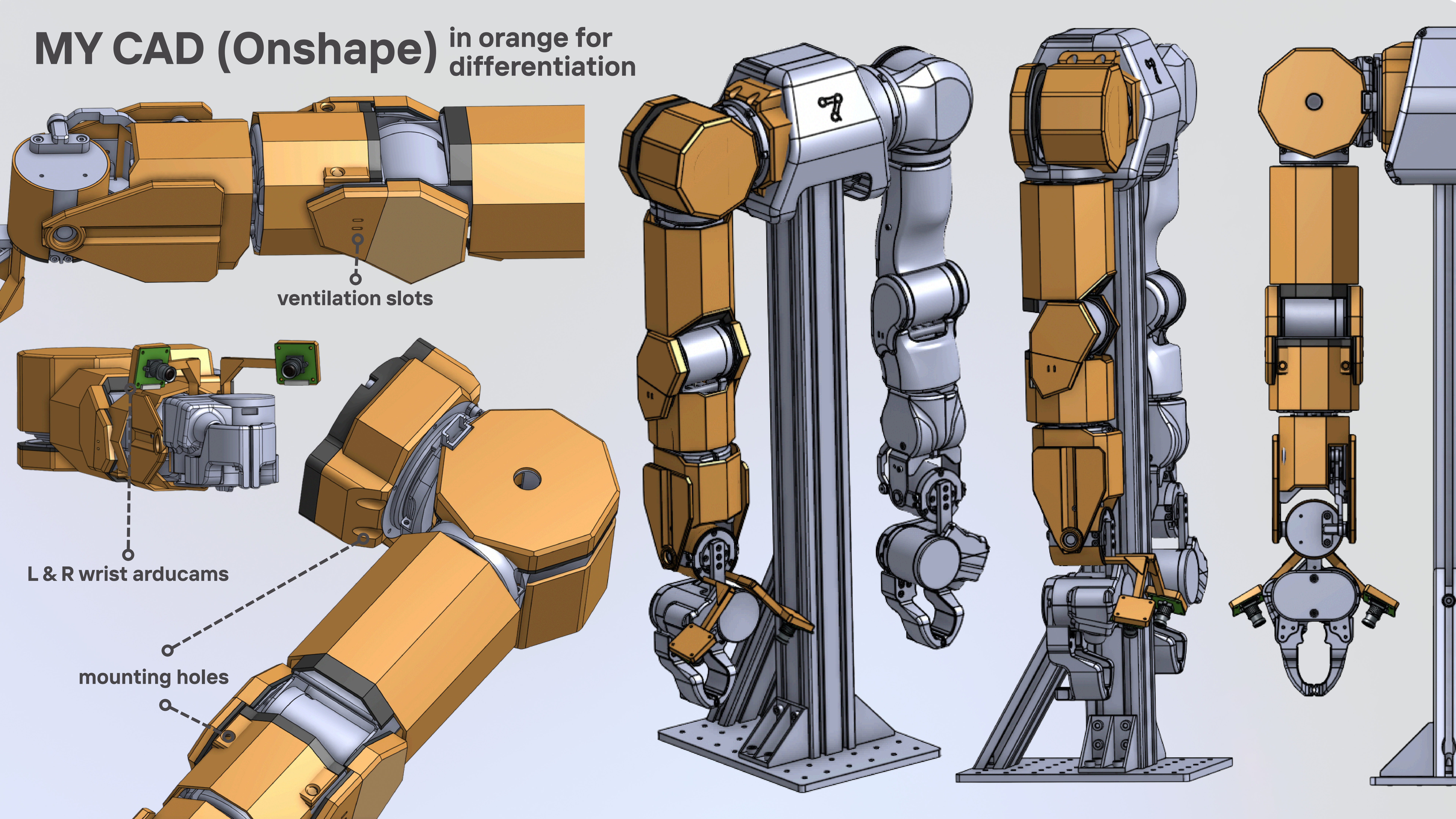
- Brushed silver thin metal panels with a lightweight engineered skin look
- Swiss-cheese perforations for weight, airflow, and exposed structure (adds CNC/finishing complexity)
- Integrated LEDs along seams to highlight joints and segmentation (extra wiring/assembly)
- More industrial, pre-production aesthetic vs PETG prototype look
- Higher manufacturing complexity (sheet metal, tighter tolerances, more finishing steps)
- Still modular and serviceable, but harder and slower to assemble

Design Direction 2: Organic Hexagonal



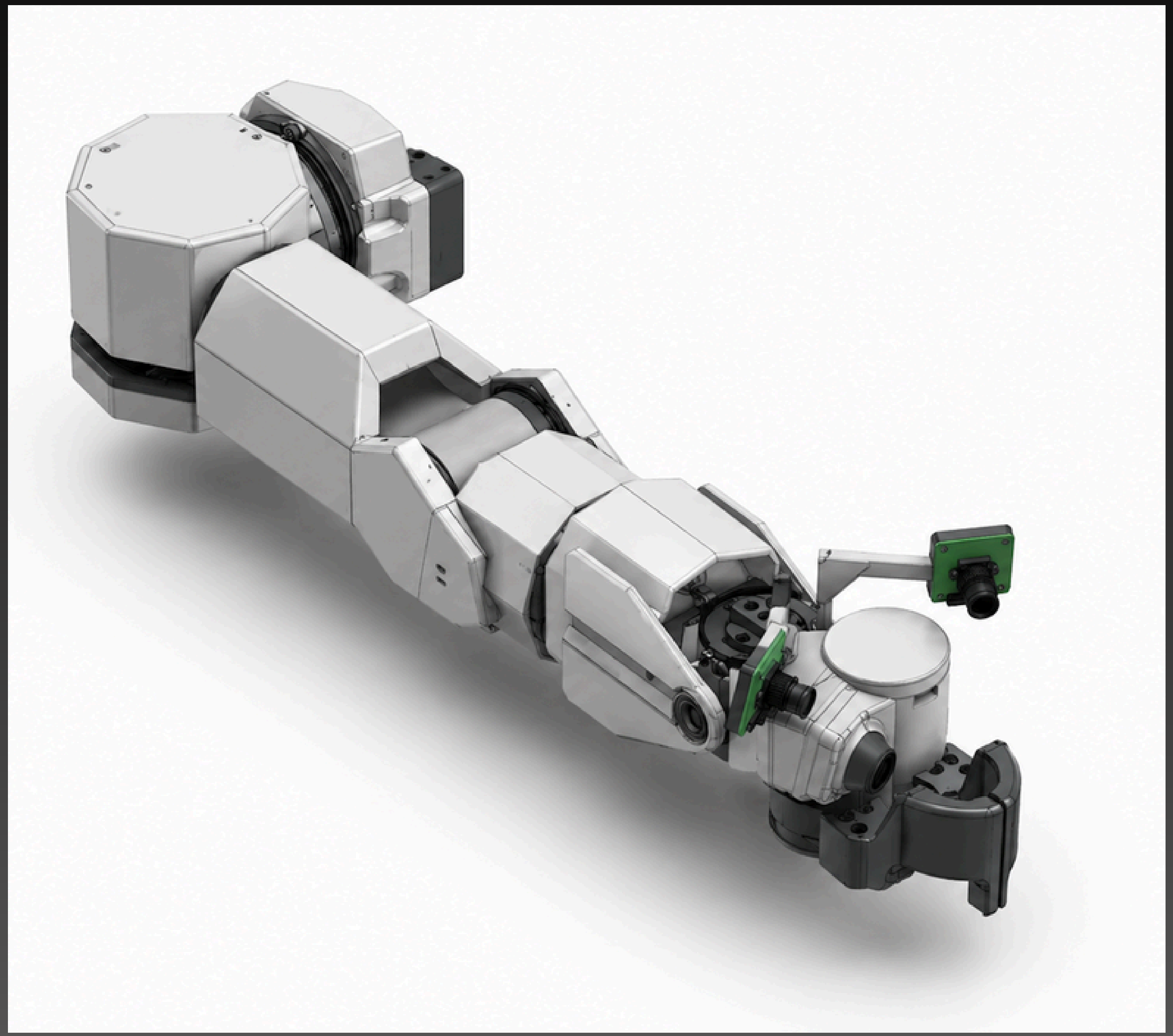


- Hand-sketched in Procreate, refined into final concept render using CAD + AI-assisted visualization
- Hexagonal geometry creates a structured, repeatable system across arm segments
- Goal: transition from exposed research robot → cohesive, intentional product design
- Matte white panels + black joint covers:
- Emphasize articulation points
- Chosen to improve visual hierarchy between static vs moving elements



MY CAD (Onshape)
in orange for differentiation
ventilation slots
L & R wrist arducams
mounting holes

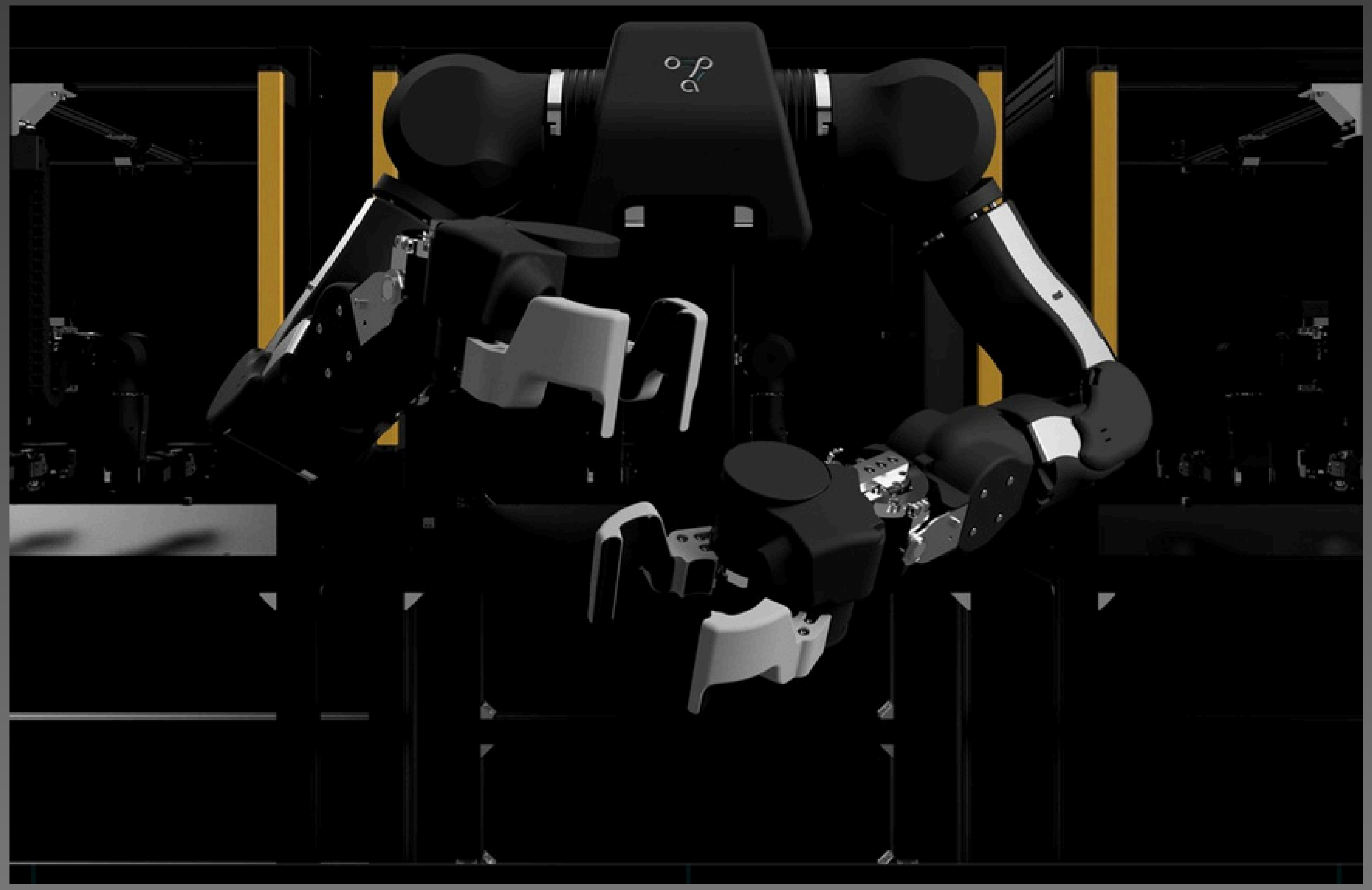
Post-CAD Render Combined with Design Direction



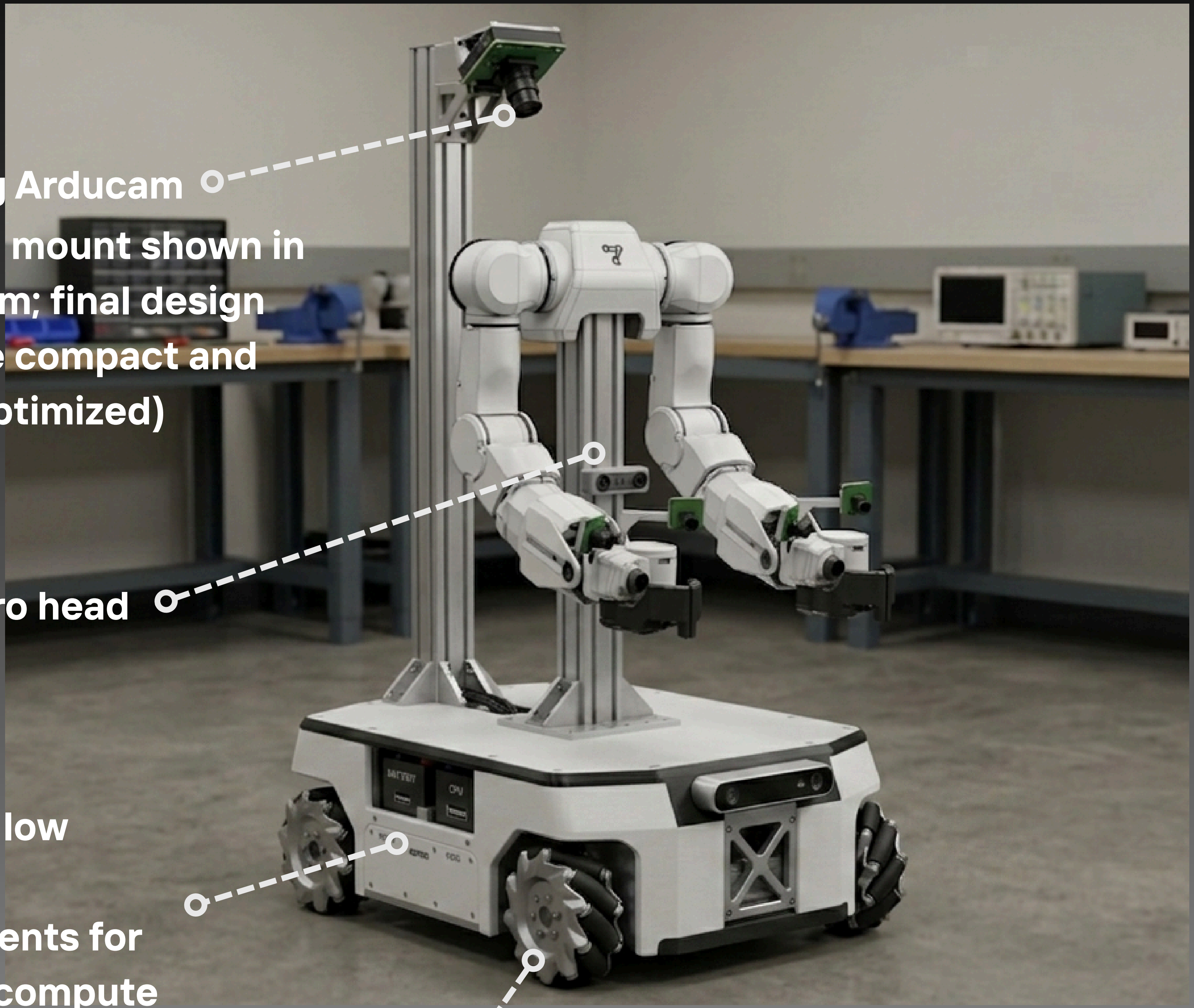
- Final CAD rendered using design direction inputs (sketch + model) to visualize full housing concept
- Includes dual Arducam wrist mounts, outer casing, and ventilation features
- Mounting holes exist in CAD, but full fastener pattern is not fully detailed
- Panel split lines are part of intended final design, not fully modeled in CAD

Cable Routing and Service Access

- Internal routing: CAN FD + camera cables use existing arm channels
- Fully internal wiring reduces snag points and maintains clean exterior profile
- Service access via segmented panels at defined split locations
- Thermal management via ventilation near actuator regions
- Panels secured through dedicated fasteners and bracket interfaces



Mobile Base



Ceiling Arducam

(Ceiling Arducam mount shown in conceptual form; final design would be more compact and weight-optimized)

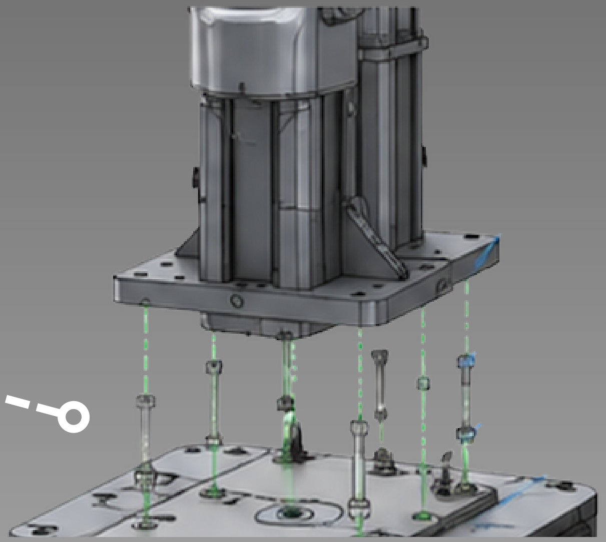
ZED stero head

Drawers & low storage compartments for batteries/compute

Omni-directional mecanum wheels

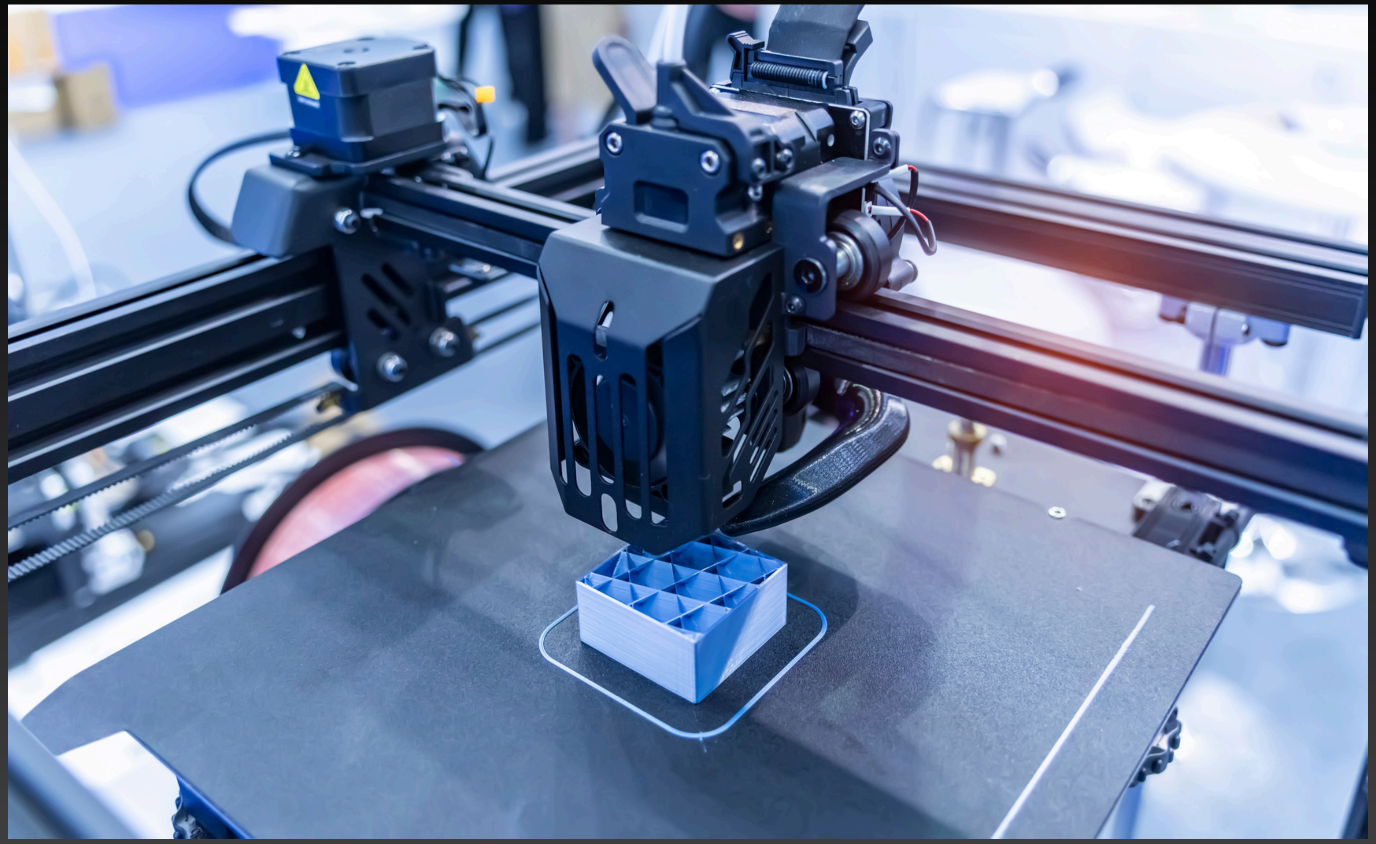
- CNC internal frame + metal chassis for load-bearing structure
- PETG exterior panels for lightweight protective shell
- Arm mounted to top plate via bolted metal interface
- Integrated drawer system for battery + compute
- Mecanum wheel base for omnidirectional motion
- Ceiling Arducam on bracketed mini-chassis, angled downward for full coverage
- ZED stereo camera centered between arm mounts at workspace height

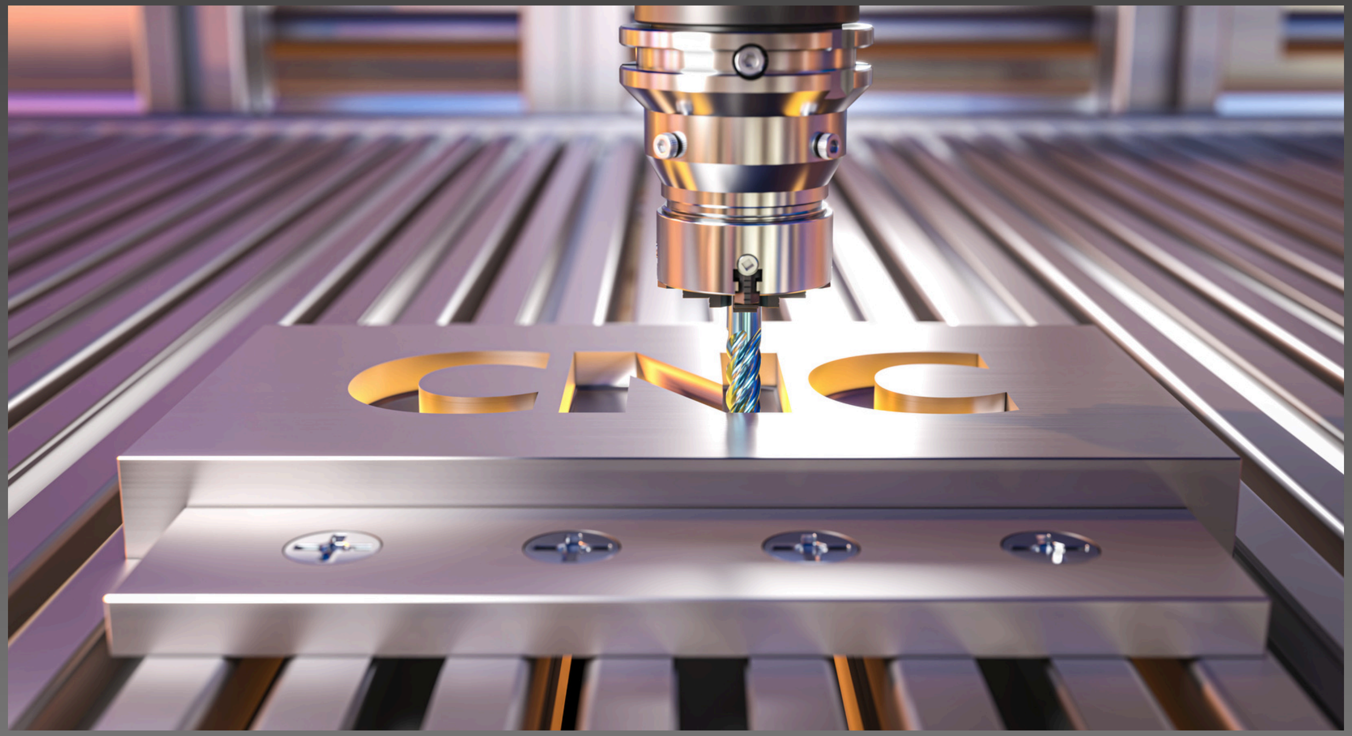
Robot post bolted into cart top plate



Manufacturability

- Panel system designed for manufacturability + iteration speed
- PETG exterior panels:
  - Heat resistance
  - Rapid prototyping
- Sheet metal reserved for internal structure + mobile base frame
- Mounting:
  - Existing M6 tap locations where available
  - 3D printed bracket interfaces where not
- Heat-set inserts used for all threaded connections
- 2.5–3 mm wall thickness for stiffness-to-weight balance
- Fully compatible with standard FDM + basic fabrication tools

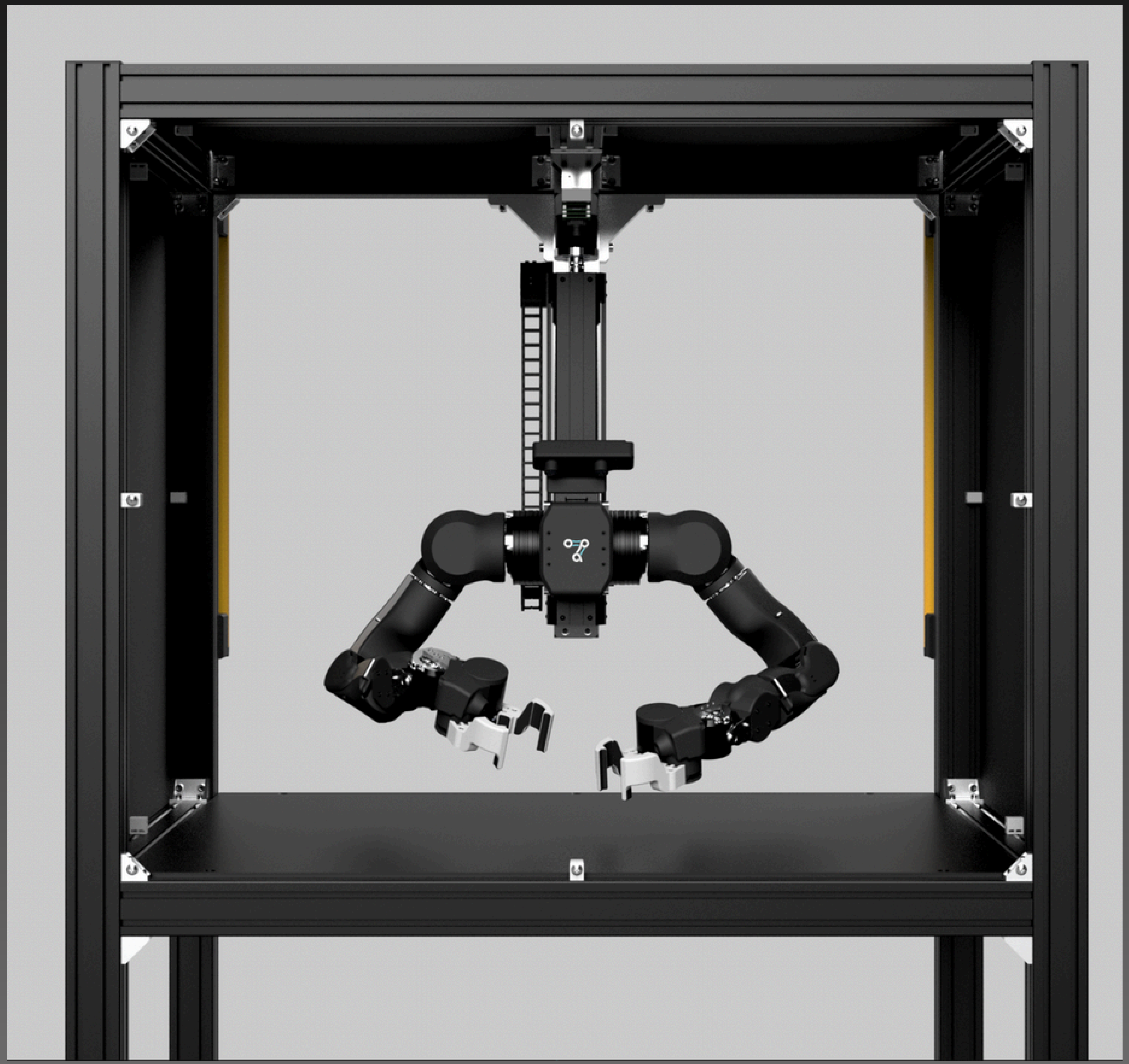




Assumptions & Design Constraints

- Internal motor/controller geometry estimated from STEP data
- Cable routing inferred from visible structure (not physically validated)
- Mounting hole alignment requires physical verification
- Camera placements outside wrist system are conceptual
- Joint clearances approximated without full motion simulation
- Panel articulation clearances require physical testing
- CAD represents continuous surfaces; final design requires defined service splits

Future Work



- Validate clearances via full motion simulation
- Refine panel split locations for service access + assembly
- Build and test physical prototypes
- Evaluate thermal performance near actuators/electronics
- Optimize weight + panel thickness for manufacturability
- Validate camera coverage across full workspace

Thank You